Naimiris A Profit-Driven Smart Farming Model for Scotch Bonnet Peppers

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Agriculture remains a critical sector in Sri Lanka, particularly in wet zone regions where many small-scale farmers depend on suitable climate and soil conditions for successful cultivation. However, many farmers still rely on traditional knowledge when planning farming activities, which can lead to inefficient resource usage, poor crop planning, and financial losses. The lack of accessible technological tools that support data-driven farming decisions further increases these challenges.

This research proposes an AI-based Decision Support System for Scotch Bonnet pepper cultivation targeting farmers and investors in the Sri Lankan wet zone. The proposed system aims to assist farmers by analyzing multiple inputs including land photos, land size, soil type, water availability, flood risk, available resources, equipment availability, number of workers, and available farming budget.

Using image analysis, weather data integration, and data-driven decision models, the system will evaluate land suitability and recommend the most appropriate farming method such as open-field farming, bag cultivation, shade-net farming, or greenhouse farming. The system will also provide recommendations on crop quantity, suitable seed varieties based on environmental conditions, and optimal planting strategies.

In addition to technical farming recommendations, the system will generate a financial overview including cost breakdown, expected yield, and return on investment (ROI). A land suitability score will also be calculated to help farmers assess the potential success of cultivation on their land.

The expected outcome of this research is a smart agricultural advisory system that supports farmers and investors in making informed farming decisions, reducing financial risks, and improving productivity within Sri Lanka’s wet zone agricultural sector.

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

ML – Machine Learning

ROI – Return on Investment

DSS – Decision Support System

IoT – Internet of Things

UI – User Interface

UX – User Experience

CNN – Convolutional Neural Network

LSTM – Long Short-Term Memory

SDG – Sustainable Development Goals

ICT – Information and Communication Technology

API – Application Programming Interface

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1. INTRODUCTION

1.1 Background and Context

Pepper is a globally important crop. In 2014, pepper covered about 3.6 million hectares worldwide (25% of people consume pepper daily), with 65% of production in Asia [1]. Brazil and Sri Lanka ranked among the top producers (54,425 t and 28,901 t in 2016, respectively)[2]. Scotch Bonnet (Capsicum Chinese) is a hot chili valued for its aroma, vitamins A/C, and flavor[3]. In Sri Lanka, Scotch Bonnet cultivation is just emerging small-scale farmers began growing hybrids only recently, often imported from Malaysia [4]. This underutilized crop has advantages (fewer pests, drought tolerance, perennial growth) that suit local conditions[5]. However, pepper yields are sensitive to climate and management, and smallholders often lack access to up-to-date agronomic advice.

Farmers (especially smallholders) are most affected. Low-income growers need higher yields and reliable production to improve livelihoods and food security. Digital tools and better decision support can help—but Sri Lanka's agriculture has faced fragmentation, data silos, and limited digital literacy. Recent government initiatives (e.g. the CROPIX digital platform) aim to unify crop data and connect farmers to planning and market information [6] [7]. These efforts are designed to strengthen food security and farmer livelihoods (SDG 2) by moving from "fragmented information to informed, timely decisions" [8] [9]. CROPIX and related ICT frameworks enable evidence-based policies, climate resilience and precision farming on a national scale [10].

Despite these advances, significant challenges remain. Key constraints include low smartphone penetration, poor rural connectivity, and high average farmer age[11]. The AFACI-RATES extension project in Sri Lanka found that "low smartphone usage, insufficient mobile network coverage, and low digital literacy among older farmers" were major barriers to digital adoption. Globally, smallholder groups face similar issues: older or less educated farmers are much less likely to use digital ag tools [12] [13]. At the same time, climate change is increasing weather volatility, making timely decisions (planting dates, irrigation, pest management) even more critical.

Existing solutions and limitations: Many existing tools address pieces of the problem. Crop-growth models (e.g. DSSAT's CSM-CROPGRO-Pepper module) can simulate yield under various conditions [14]. IoT devices and AI (e.g. CNNs, drones, sensors) have been deployed

for specific tasks like disease detection or irrigation scheduling [15] [16]. For example, AI-based edge solutions have been used to detect chili pest incidence in real time [17] [18]. However, most systems target broad regions or controlled environments (greenhouses), and they often require stable internet, expensive hardware, or technical expertise. In Sri Lanka, digital advisory services (via WhatsApp or mobile apps) have improved knowledge access, but their impact is limited by the above barriers.

Proposed solution idea: We propose an integrated AI-driven decision support system specifically for Scotch Bonnet cultivation. The system will gather multi-modal data (farmland images, climate, soil tests, farmer resources) via smartphone or local sensors, then apply AI/ML (computer vision, predictive modeling) to analyze land suitability and crop status [19]. Based on this analysis and resource constraints (land, budget), it will recommend optimal practices (e.g. variety choice, planting schedule, pest control, irrigation) tailored to each farmer. By combining image analysis with local data, the system aims to be user-friendly and robust to connectivity issues (processing can occur on-device or via lightweight IoT) [20]. This approach aligns with SDG 2 (“zero hunger”) by helping double smallholder productivity [21]and with SDG 12 (sustainable production) by reducing waste through precision management. It falls under research clusters like AI/Machine Learning and Data Analytics, contributing novels, locally-adapted attach.

1.2 Literature Review

Recent studies highlight the potential of AI/IoT in precision agriculture. Reviews show that integrating computer vision and sensor networks can significantly improve crop monitoring and management [9]. For example, convolutional neural networks (CNNs) have enabled early disease detection in crops, while LSTM models help predict yields and weather impacts, all supporting data-driven farm decisions [12]. Smart farming research covers tasks like soil moisture sensing, pest detection and nutrient optimization. Uchenna et al. (2025) survey IoT in smart agriculture, noting how sensor-actuator networks allow “real-time decision making and precise control” of irrigation, fertilization and environment [22]. Computer-vision surveys similarly stress the role of drones and cameras in yield estimation and land analysis [16].

Decision Support Systems (DSS): The literature on agricultural DSS shows a shift from manual to digital tools. Zhai et al. (2020) note that Agri 4.0 (IoT + AI) enables collection of "enormous farming data" to guide farmers toward higher profits [23]. Classic crop models (e.g. DSSAT) can predict pepper growth under varying climate [6]. However, modern DSS increasingly incorporate machine learning: they adapt to current weather and market data for real-time guidance [4]. An emerging trend is AI-chatbot advisors. Asolo et al. (2024) developed an AI-chatbot DSS that gives farmers personalized advice via natural language [20].. This study (in J. Digital Food Energy Water Syst.) shows such systems can boost productivity and cut environmental impact [20].It underscores that farmer acceptance depends on trust, simplicity and local relevance, not just tech sophistication.

Pepper-specific research: Though much agri-tech is crop-agnostic, there are some pepper-focused works. Nautiyal et al. (2025) review AI in agriculture and mention models for yield and pest control [18] [24]. Kim et al. (2023) built an edge-AI system for chili (a Capsicum spp.) pest detection on small farms, noting the special challenge of unstructured, open-field environments. These solutions reduced latency and cost for low-resource farmers [25]. A recent Sri Lankan study (Charana et al., 2025) characterized local Scotch Bonnet varieties and found they thrive with fewer pests and drought compared to common peppers. However, that work also points out "C. Chinese remains underutilized in commercial agriculture in Sri Lanka", implying a need for better promotion and management tools.

Existing products/services: Some commercial platforms aim to connect farmers to data. Sri Lanka's CROPIX (2026) is a national DPI linking farmers, extension officers and policymakers around a single crop-data platform. The system promises to "move from fragmented information to informed, timely decisions" for food security. Globally, apps like Cropping or Farm Beats offer yield forecasts and irrigation advice, but they often require steady connectivity or focus on major row crops. In practice, many farmers still rely on extension hotlines or SMS alerts; these have improved basic yields but lack sophistication.

1.3 Research Gap

Despite progress in agricultural AI, we see clear gaps. Existing DSS and models rarely address the whole system of small-scale pepper farming. Many approaches target single issues (e.g. greenhouse climate control or one crop disease) [26], and they assume laboratory conditions or large farms. In open-field scenarios, AI adoption remains very low: one study notes only ~11% of fields use AI tech, due to variable weather, unreliable networks and high costs [18]. Likewise, mobile extension pilots in Sri Lanka have boosted information access, but "low smartphone usage, insufficient network coverage, and low digital literacy among older farmers" still limit impact [11]. This suggests that a new solution must be both technically and socially tailored.

Concretely, no solution currently provides personalized, actionable advice to Scotch Bonnet growers based on local data inputs. While soil and plant models exist, they are not packaged into a farmer-friendly system for this crop. Similarly, AI-chatbot research [20] is promising but needs real-world testing in the field. Therefore, our proposed system will be novel and creative by combining multiple data streams (imagery, climate, budgets) and by delivering recommendations via an intuitive interface (e.g. smartphone chatbot or simplified dashboards). In evidence, literature underscores this gap: smallholder-focused studies call for holistic frameworks that span from government policy to field-level tech [12]. By focusing on a high-value but under-served crop (Scotch Bonnet) and by addressing known barriers (connectivity, user skill, cost), this project aims to fill a critical niche in Agri-AI research.

Table -0-1 - comparison of another research

	Research A – Sri Lanka University (2021) Crop Recommendation System [2]	Research B – Tamil Nadu Agricultural University (2022) Farm Advisory Tool [3]	Research C – Scoping Review of Agricultural DSS (2023) [4]	Research D – Subasinghe et al. (2023) Habanero DSS [5]	Proposed System AI-Based Smart Decision Support System
Farming Method Recommendation	✗	✗	✗	✗	✓
Crop Quantity Recommendation	✗	✗	✗	✗	✓
Seed Variety Recommendation	✗	✓	✓	✓	✓
Financial & Production Analysis	✗	✓	✓	✓	✓
Land Suitability Score	✗	✗	✗	✗	✓

Table 1 Comparisons of the Proposed System with Existing Agricultural Decision Support Systems

2. OBJECTIVE

2.1 Main Objective

The primary objective of this research is to design and develop an advanced, AI-driven system capable of identifying, analyzing, and mitigating procrastination behavior among students in digital learning environments. With the increasing reliance on online education platforms, students often struggle with time management and self-discipline, leading to delayed task completion and reduced academic performance. This research aims to address this issue by creating a system that continuously monitors student engagement and learning patterns, enabling early detection of procrastination tendencies. By leveraging intelligent algorithms, the system will transform raw behavioral data into meaningful insights that can support both students and educators in improving learning outcomes.

2.2 Specific Objectives

A key objective of the proposed system is to accurately identify procrastination patterns by analyzing various student interaction metrics such as login frequency, duration of study sessions, intervals between activities, and assignment submission timelines. These behavioral indicators will allow the system to detect irregularities and deviations from consistent learning habits. In addition, the system seeks to develop a robust predictive model using machine learning techniques to classify students based on their level of procrastination risk. This predictive capability will enable proactive intervention rather than reactive solutions, thereby significantly improving the effectiveness of the system.

Another important objective is to establish individualized baseline learning profiles for each student. Instead of applying a generalized model, the system will learn the normal study behavior of each user and use this as a reference point to detect abnormal patterns. This personalization ensures higher accuracy and relevance in predictions. Furthermore, the system aims to implement adaptive active time learning mechanisms, which dynamically recommend optimal study schedules based on user behavior. These recommendations will help students utilize their most productive time periods effectively, reducing the likelihood of procrastination.

The system also intends to incorporate context-aware assessment by considering external factors such as assignment deadlines, workload intensity, and timing of activities. This allows

the system to provide more realistic and situationally appropriate evaluations of student behavior. Additionally, the proposed solution will generate theory-based explanations and actionable recommendations grounded in established educational and psychological principles, particularly self-regulated learning theories. By doing so, the system not only identifies problems but also educates users on how to improve their learning habits. Ultimately, the overall objective is to enhance student engagement, promote disciplined study behavior, and improve academic performance through intelligent, personalized, and adaptive interventions

3. METHODOLOGY

The methodology of this research is designed to systematically develop and evaluate an intelligent procrastination detection and intervention system using a combination of data-driven and machine learning approaches. The process begins with data collection from various sources within online learning environments, such as learning management systems and user interaction logs. The collected data will include critical behavioral indicators such as login frequency, time spent on learning materials, assignment submission timestamps, navigation patterns, and periods of inactivity. This comprehensive dataset will provide a strong foundation for analyzing student behavior and identifying procrastination patterns.

Following data collection, a crucial step in the methodology is data preprocessing, which ensures the quality, consistency, and reliability of the dataset. This involves handling missing or incomplete data, removing inconsistencies, and normalizing features to ensure uniformity across different data types. Time-based data will be carefully processed to extract meaningful temporal patterns, such as delays in task completion and irregular study intervals. This stage also includes data transformation and structuring, preparing the dataset for effective use in machine learning models.

Feature engineering plays a significant role in enhancing the performance of the predictive system. In this stage, relevant features that directly reflect procrastination behavior will be extracted and constructed from the raw data. Examples include the frequency of last-minute submissions, duration of inactive periods, variation in study session lengths, and consistency of learning routines. These features will provide deeper insights into student behavior and improve the accuracy of the prediction models. Careful selection of features ensures that the system focuses on the most impactful indicators of procrastination.

The next phase involves the development of machine learning models to predict and classify procrastination behavior. Various algorithms such as decision trees, random forests, and logistic regression will be explored to determine the most effective approach for the given dataset. These models will be trained using historical data and evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score. The goal is to develop a reliable model that can accurately identify students at risk of procrastination and categorize them into different risk levels, enabling targeted interventions.

In addition to prediction, the methodology includes the implementation of an adaptive learning mechanism that continuously analyzes user behavior and provides personalized

recommendations. This mechanism will generate suggestions such as optimal study times, reminders for upcoming deadlines, and strategies to improve time management. The system will also adjust its recommendations dynamically based on changes in user behavior, ensuring continuous improvement and adaptability. This personalized approach is essential for effectively addressing procrastination, as each student has unique learning habits and challenges.

The system development phase involves building a fully functional web-based platform that integrates all components, including data processing, prediction models, and user interaction interfaces. The backend will handle data storage, processing, and model execution, while the frontend will provide an intuitive interface for users to view insights, recommendations, and progress. Finally, the system will undergo thorough evaluation and testing to assess its performance, usability, and effectiveness. Both quantitative metrics and qualitative user feedback will be used to validate the system, ensuring that it meets its objectives and provides real value in reducing procrastination and improving academic performance

4. HIGH-LEVEL SYSTEM ARCHITECTURE

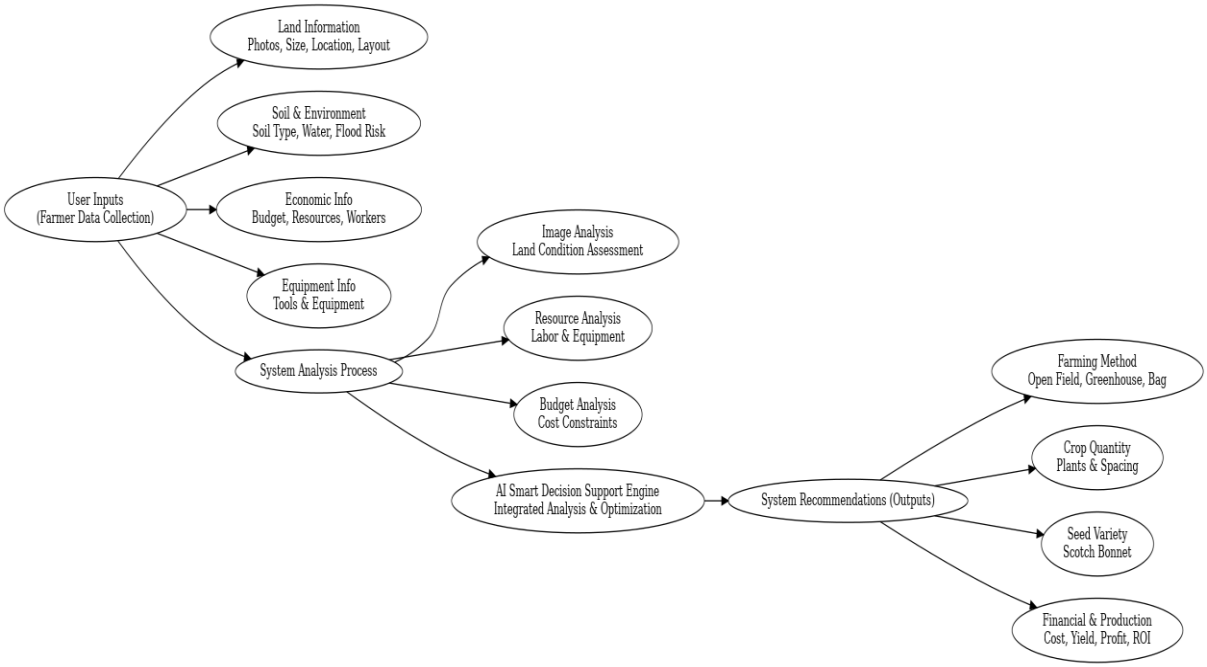


Figure -0-1 - System diagram

The proposed system is designed using a comprehensive and modular architecture that enables efficient data flow, intelligent analysis, and user-centered interaction. The architecture begins with the data input layer, which is responsible for collecting both real-time and historical data from various sources such as learning management systems and user activity logs. This layer ensures that all relevant behavioral data, including study patterns, interaction frequency, and task completion timelines, are continuously captured and fed into the system for analysis.

Once the data is collected, it is processed in the data processing layer, where it undergoes cleaning, transformation, and feature extraction. This layer plays a critical role in converting raw, unstructured data into meaningful and structured formats suitable for analysis. It ensures that inconsistencies are removed, missing values are handled appropriately, and relevant behavioral features are extracted. By preparing high-quality data, this layer directly contributes to the accuracy and reliability of the system's predictive capabilities.

At the core of the architecture lies the prediction engine, which utilizes machine learning models to analyze processed data and identify patterns associated with procrastination behavior. This component is responsible for generating predictions regarding the likelihood and severity of procrastination for each user. It continuously updates its predictions based on new data, allowing the system to adapt to changes in user behavior over time. This dynamic capability ensures that the system remains accurate and relevant in real-world usage.

Supporting the prediction engine is the user profiling module, which maintains detailed records of individual user behavior and learning patterns. This module creates and updates baseline profiles for each student, enabling the system to compare current behavior against historical norms. By incorporating personalized data, the system can provide more accurate and meaningful insights, as it accounts for individual differences rather than relying on generalized assumptions.

The adaptive recommendation engine is another key component of the architecture, responsible for translating predictions into actionable interventions. Based on the identified procrastination risk, this module generates personalized recommendations such as optimized study schedules, reminders, and productivity strategies. These recommendations are designed to guide students toward better time management and more consistent learning habits. Additionally, the system includes a context-aware analysis module that enhances decision-making by considering external factors such as deadlines, workload, and timing, ensuring that recommendations are practical and situation-specific.

The user interface layer serves as the point of interaction between the system and its users, providing a clear and intuitive platform for accessing insights, recommendations, and progress reports. It is designed to be user-friendly, enabling students and educators to easily understand and utilize the system's outputs. Finally, the database layer securely stores all system data, including user profiles, behavioral records, and prediction results, ensuring efficient data management and retrieval. Together, these interconnected components form a robust and scalable architecture capable of effectively addressing procrastination in online learning environments.

5. USER REQUIREMENTS

The proposed system is designed to address the practical and economic needs of two primary user groups; namely small-scale farmers engaged in Naimiris cultivation and potential investors who are interested in generating income through agricultural ventures. Small-scale farmers often operate under significant constraints, including limited access to modern agricultural knowledge, lack of technical decision-making support, unpredictable weather conditions, and unstable market opportunities. As a result, the system must provide a highly intuitive and accessible interface that allows farmers to easily interact with the platform regardless of their technical background. The system should support mobile-based access and local language options to ensure inclusivity, particularly for users in rural areas. Farmers should be able to input simple parameters such as land size, soil type, location, and available resources, after which the system will generate clear, actionable recommendations regarding suitable cultivation methods, optimal planting periods, irrigation strategies, and expected yield outcomes. Furthermore, the system must provide real-time alerts, reminders, and adaptive suggestions to help farmers improve productivity and reduce risks associated with poor decision-making.

In addition to farmers, investors represent a key stakeholder group whose requirements differ in terms of analytical depth and financial focus. Investors require accurate, data-driven insights that can support informed decision-making and risk assessment. Therefore, the system must provide advanced features such as land suitability analysis, predicted yield outputs, cost-benefit evaluations, and return on investment (ROI) estimations. Investors should also be able to explore different farming scenarios, compare potential outcomes, and identify profitable opportunities within Naimiris cultivation. The system should present this information through clear visualizations and structured reports, ensuring transparency and ease of understanding. Additionally, the platform should facilitate potential collaboration between farmers and investors, creating a mutually beneficial ecosystem where resources, knowledge, and financial support can be shared effectively. Overall, the system must ensure reliability, scalability, and personalization to meet the diverse needs of its users while promoting sustainable agricultural development and economic growth.

Table -0-2 - Users

User Type	**Key Needs**	**System Support Features**
Small-Scale Farmers	Easy-to-use system, farming guidance, yield prediction, risk reduction	Mobile-friendly UI, local language support, crop recommendations, alerts, reminders
Investors	Profit analysis, ROI estimation, land suitability insights, risk assessment	Data analytics dashboard, financial reports, predictive modeling, visualization tools

Table -3 - Stakeholder

Stakeholder	**Role in System**	**Benefit**
Farmers	Primary users	Increase yield, reduce risk, improve income
Investors	Financial contributors	Maximize profit, minimize investment risk
Agricultural Experts	Knowledge providers	Share expertise, improve farming practices
Government / Agencies	Regulatory and support bodies	Promote sustainable agriculture and rural development
System Developers	System designers and maintainers	Ensure performance, scalability, and system improvement

6. COMMERCIALIZATION PLAN

The commercialization strategy for the proposed system focuses on transforming the developed solution into a scalable and sustainable agri-tech platform that delivers economic value to both farmers and investors. Initially, the system will be introduced as a digital platform accessible through both web and mobile applications, ensuring maximum reach and ease of adoption across rural and urban regions. To encourage widespread usage among small-scale farmers, a freemium business model will be implemented, where basic features such as general crop recommendations, simple predictions, and alerts are provided free of charge. This approach helps to lower entry barriers and build user trust. Advanced features, including detailed analytics, personalized advisory services, and investment-related insights, will be offered through a subscription-based model, generating a steady revenue stream for the platform.

For investors, premium services will include comprehensive financial analysis, land suitability reports, yield forecasting, and risk evaluation tools. These services will enable investors to make well-informed decisions and maximize returns on their investments. The platform will also act as a bridge between farmers and investors by facilitating partnerships, allowing investors to fund agricultural projects while farmers benefit from financial support and improved resources. Strategic collaborations with agricultural organizations, government bodies, and cooperatives will further enhance the system's credibility and adoption. Additionally, partnerships with supply chain stakeholders such as distributors and exporters can open new market opportunities for farmers, ensuring better pricing and income stability.

Marketing and expansion strategies will include awareness campaigns, farmer training programs, and digital outreach through social media and agricultural networks. Over time, the system can be expanded to support additional crops and geographical regions, increasing its commercial viability and impact. By integrating technology with agriculture and finance, the proposed solution has strong potential to become a profitable venture while contributing to sustainable development and rural economic empowerment.

7. BUDGET AND JUSTIFICATION

The budget for the proposed system has been carefully structured to cover all essential aspects required for successful development, deployment, and long-term operation. The primary cost component is system development, which includes frontend and backend development, user interface design, and the implementation of machine learning models. This phase requires skilled technical resources and forms the foundation of the entire system. Another significant cost involves data collection and processing, which includes acquiring agricultural, environmental, and market data, as well as cleaning and preparing this data for analysis. High-quality data is critical to ensuring accurate predictions and reliable system performance.

Infrastructure costs are also a key consideration, as the system will require cloud-based hosting services, database management systems, and server resources to handle data storage and processing efficiently. These components ensure scalability and allow the system to support a growing number of users. Testing and deployment costs are allocated to ensure that the system functions correctly under various conditions, including debugging, performance optimization, and final deployment. Additionally, ongoing maintenance costs are necessary to support system updates, performance monitoring, and user assistance, ensuring that the platform remains reliable and up to date.

The budget is justified by the significant value the system provides to its users. For farmers, the system helps improve productivity, reduce losses, and increase income through better decision-making. For investors, it minimizes financial risks and enhances profitability through accurate forecasting and analysis. By enabling efficient resource utilization and promoting data-driven agriculture, the system offers strong long-term returns that outweigh the initial investment costs. Therefore, the proposed budget is both reasonable and essential for achieving the objectives of the project.

Category	Description	Estimated Cost (LKR)
System Development	Self-development (you + team), simple frontend/backend	40,000
Data Collection	Manual data + free datasets	5,000
Infrastructure	Free tier (Firebase / GitHub / free cloud)	10,000
Testing & Deployment	Basic testing (no paid tools)	5,000
Maintenance	Short-term (self handling)	10,000
Miscellaneous	Printing, small expenses	5,000
Total		**75,000**

Figure 0-2 - Budget

8. WORK BREAKDOWN STRUCTURE

The Work Breakdown Structure (WBS) of this project organizes the development of the AI-based Smart Decision Support System into a series of well-defined phases aligned with the project timeline and system components. The structure ensures that all activities, from initial planning to final delivery, are systematically managed and completed within the given schedule. It also supports balanced workload distribution among team members while maintaining a clear connection to the Gantt chart milestones.

The project begins with the problem identification and planning phase, where the research problem related to farming challenges in the Sri Lankan wet zone is analyzed. During this stage, system objectives are defined, initial requirements are gathered, and the project proposal is prepared and submitted for approval. This phase establishes the foundation for the entire system by clearly identifying the scope and expected outcomes.

Following this, the literature review phase focuses on studying existing smart farming solutions and AI-based decision support systems. This involves reviewing scholarly articles, analyzing current technologies, and identifying limitations in existing approaches. The purpose of this phase is to highlight the research gap and justify the need for the proposed system.

The system design phase translates the identified requirements into a structured solution. It includes designing the overall system architecture, defining key modules such as user input, analysis, AI decision engine, and recommendation system, as well as preparing database designs and user interface layouts. This phase ensures that all components are logically organized before development begins.

Next, the development process starts with the implementation of the data collection module. This module allows users to input essential information such as land details, images, soil data, and available resources. The system ensures proper validation and structured data handling to support accurate analysis in later stages.

The project then proceeds to the analysis phase, where the system processes the collected data. Image analysis is performed to evaluate land conditions and suitability, while resource and budget analysis determine the scale and feasibility of farming activities. These analytical processes prepare the necessary inputs for intelligent decision-making.Testing and validation follow integration, where the system undergoes unit testing, system testing, and user

acceptance testing. Performance and accuracy are evaluated to ensure that the system meets the required standards and produces reliable results.

Finally, the project concludes with documentation and final deliverables. This includes preparing the final report, completing technical documentation, deploying the system, and conducting the final presentation and viva. The research paper submission and publication evidence are also completed during this stage.

Overall, the WBS provides a structured approach to developing the proposed system by breaking down the project into manageable phases. It ensures efficient task allocation, alignment with the project timeline, and successful achievement of all project objectives.

9. GANTT CHART

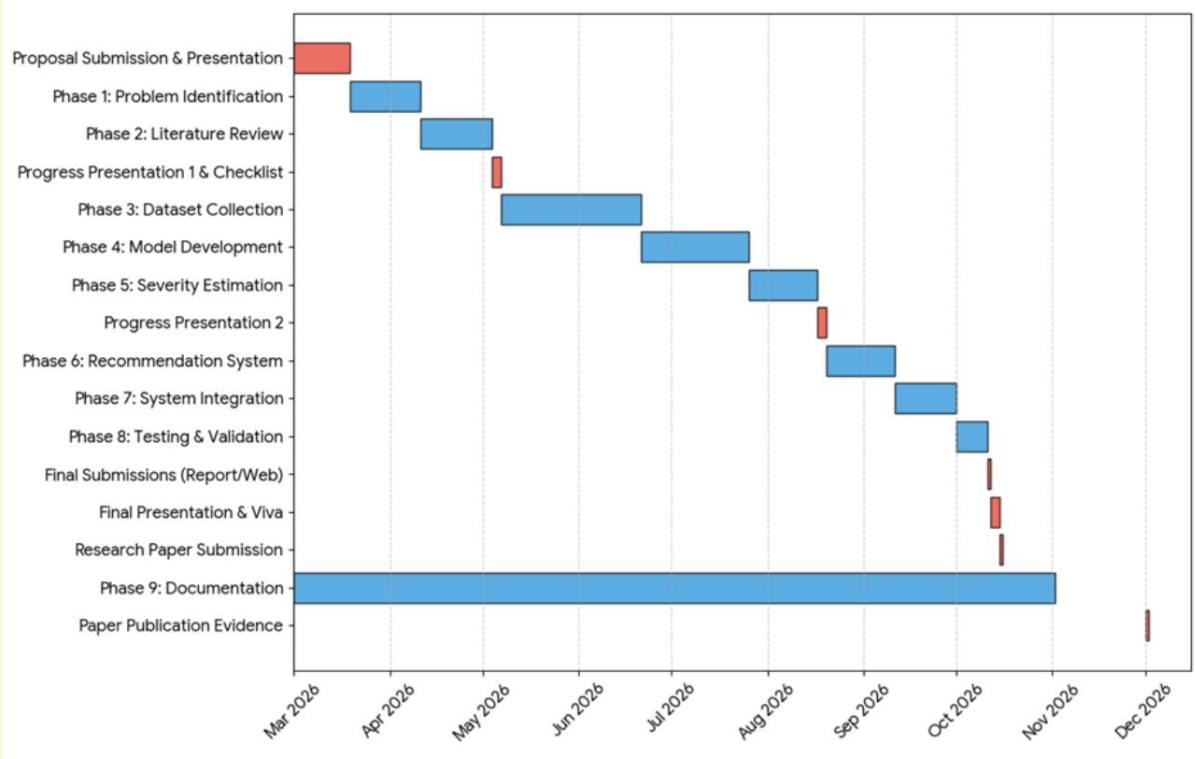


Figure -3 - GANTT CHART

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11.APPENDICES

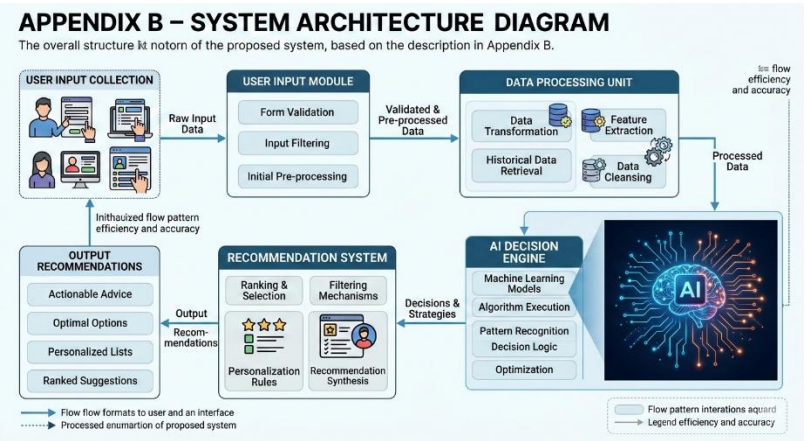
Appendix A – Survey Questionnaire

The following questionnaire was designed to collect information from farmers and potential users to understand their challenges, resource availability, and expectations from the proposed system.

Sample Questions:

1. What type of land do you use for cultivation? (Flat / Rooftop / Other)
2. What is the size of your land?
3. What type of soil do you have? (Clay / Sandy / Loamy)
4. What is your average farming budget?
5. Do you currently use any technology for farming decisions? (Yes/No)
6. What are the main challenges you face in farming?
7. How many workers are available for your farming activities?
8. What type of equipment do you currently use?
9. Are you interested in AI-based farming recommendations? (Yes/No)
10. What kind of support do you expect from a smart farming system?

Appendix B – System Architecture Diagram



Appendix C – User Interface Mockups

The user interface mockups provide a visual representation of how users will interact with the system. The design focuses on simplicity and usability to ensure accessibility for farmers with limited technical knowledge. Key interfaces include the data input screen, dashboard view, recommendation display, and report generation page. The mockups demonstrate how information is presented clearly and how users can easily navigate through the system.

Appendix D – Risk Analysis

The following table identifies potential risks associated with the system along with their impact and mitigation strategies.

Risk	Description	Impact	Mitigation Strategy
Data Inaccuracy	Incorrect user input or poor data quality	High	Implement input validation and data checks
Technical Failure	System errors or crashes	Medium	Regular testing and backup systems
Low User Adoption	Farmers may not use the system	High	Provide simple UI and user training
Connectivity Issues	Poor internet access in rural areas	High	Enable offline features or lightweight processing
Security Risks	Data privacy and unauthorized access	High	Use encryption and secure authentication

Appendix E – Ethical Considerations and Data Privacy

The proposed system ensures that all user data is handled responsibly and ethically. Personal and farming-related data collected from users will be used only for system analysis and recommendation purposes. Proper security measures such as data encryption, secure authentication, and controlled access will be implemented to protect user information. Users will be informed about how their data is used, and their consent will be obtained before data collection. Additionally, the system will avoid any bias in decision-making and ensure fairness and transparency in all recommendations. These practices help maintain user trust and comply with ethical standards in technology development.